



Wide-angle frames

with distributed AprilTag boards

Detect Boards and Extract Outer4

board IDs and outer-corner observations

Outer4 Bootstrap

camera model + temporary poses

intermediate model

Recover Internal Observations

spherical interpolation
+ local image refinement

internal observations

Outer4 Observations

kept for final calibration

Combined Observation Set

$$\mathcal{O} = \mathcal{O}^{\text{out}} \cup \mathcal{O}^{\text{int}}$$

Multi-board Bundle Adjustment

camera intrinsics · inter-board geometry · frame poses